HWA CHONG INSTITUTION
JC2 Preliminary Examinations
Higher 2

CANDIDATE NAME		**CT GROUP**	15S ___
CENTRE NUMBER		**INDEX NUMBER**	

BIOLOGY — 9648 / 03

Paper 3 Applications Paper and Planning Question — **16 September 2016**

Additional Materials: Writing Paper — **2 hours**

INSTRUCTIONS TO CANDIDATES

Write your **name**, **CT group**, **Centre number** and **index number** in the spaces provided at the top of this cover page.

STRUCTURED QUESTIONS

Answer **all** three questions.

Write your answers on the lines / in the spaces provided.

PLANNING QUESTION

Answer the question in booklet **4**.

Write your answers on the lines / in the spaces provided.

FREE RESPONSE QUESTION

Answer the question.

Your answers must be in continuous prose, where appropriate.

Write your answers on the writing paper provided.

BEGIN EACH PART ON A FRESH SHEET OF WRITING PAPER.

A **NIL RETURN** is required for parts not answered.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [ ] at the end of each question or part question.

The use of an approved scientific calculator is expected, where appropriate. You may lose marks if you do not show your working or if you do not use appropriate units.

You are reminded of the need for good English and clear presentation in your answers.

For Examiners' Use	
Question	**Marks**
1	**/ 13**
2	**/ 12**
3	**/ 15**
4	**/ 12**
5	**/ 20**
Total	**/ 72**

BOOKLET 1

STRUCTURED QUESTIONS

QUESTION 1

Bacteria can be genetically modified to produce insulin for human use. To achieve this, human insulin genes are transferred into bacteria. Plasmids containing two antibiotic resistance genes, one coding for resistance to tetracycline and one for resistance to ampicillin, are used to carry out this transfer.

A restriction enzyme was used to cut the human DNA and plasmids. Fig. 1.1 shows the different human DNA fragments and the linearised plasmid that was produced.

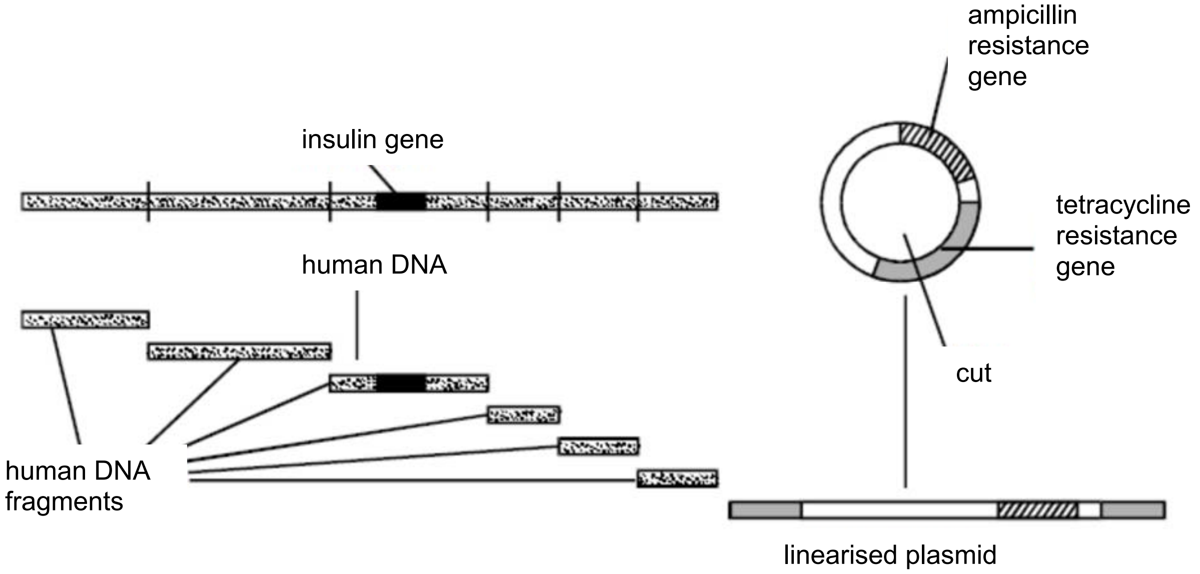

Fig. 1.1

(a) Describe how the restriction enzyme cuts the human DNA and plasmid.

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The human DNA fragments and linearised plasmids were mixed together with DNA ligase. The linearised plasmids are then mixed with bacteria. Some of the bacteria take up the plasmids.

(b) **(i)** Explain how it is possible to distinguish between bacteria which have taken up a plasmid containing a human DNA fragment and those which have taken a plasmid without any extra DNA.

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(ii) Suggest why genes for antibiotic resistance are now rarely used as selectable markers in genetic engineering.

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Unlike diabetes, Huntington's disease (HD), an autosomal dominant disorder, has a clear genetic basis. The gene involved contains a section of DNA with many repeats of the base sequence CAG. The number of repeats found in an allele of this gene may determine how early an individual develops HD. In genetic testing for HD, fragments of DNA are cut from an individual's alleles and separated by gel electrophoresis to determine their length.

Four members of a family affected by HD were tested:

- **A** is the father of **B** and developed symptoms of HD in old age.
- **B** is the mother of **D** and developed symptoms of HD in middle age.
- **C** is the **unaffected** father of **D**.
- **D** is the son of **B** and **C** and developed symptoms of HD as a child.

The results of the genetic test are shown in Fig. 1.2.

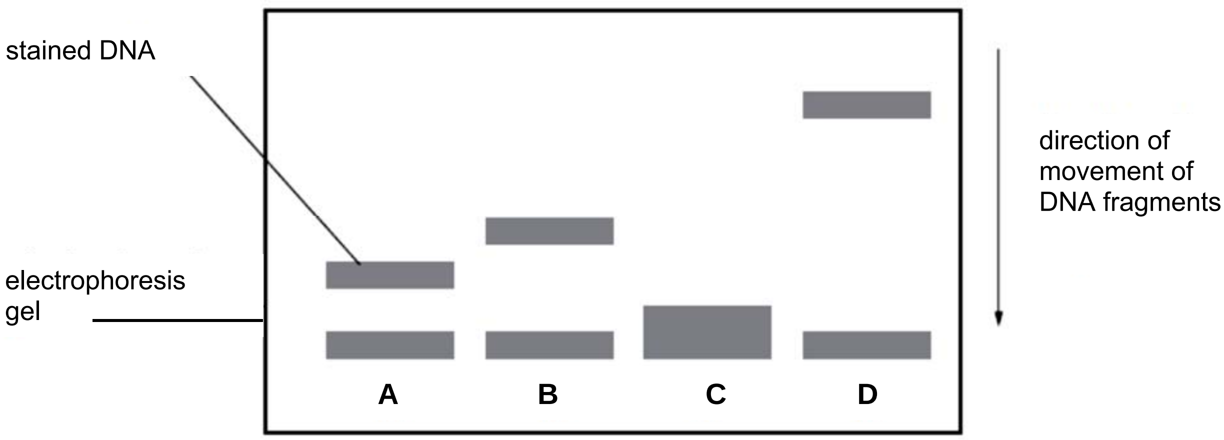

Fig. 1.2

(c) With reference to Fig. 1.2,

(i) describe the relationship between length of DNA fragment and age of onset of HD.

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(ii) explain the difference in positions of DNA fragments from the different members of the family in the test.

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[Total: 13]

QUESTION 2

Duchenne Muscular Dystrophy (DMD) is a progressive muscle wasting disease caused by mutation in the *dmd* gene. The *dmd* gene is the largest gene in humans and it encodes for the dystrophin protein. Mutations in the *dmd* gene result in a nonfunctional dystrophin protein, hence resulting in the disease.

Since the discovery of the *dmd* gene, investigations have been made to exploit gene therapy as a possible cure for DMD. In past studies, DNA library was produced by splicing together essential exons. This reduces the size of the *dmd* gene, allowing viral vectors to carry it.

(a) Identify the type DNA library and outline how the DNA library is produced.

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(b) Explain the limitations of using viruses in the gene transfer of the *dmd* gene.

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(c) Normal muscle fibres can be generated if adult mesenchymal stem cells, without the genetic defect that causes DMD, is delivered into the patients' muscles. Mesenchymal stem cells can give rise to many types of cells including bone, cartilage, lung and muscle cells.

Describe the unique features of adult stem cells.

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Fig. 2.1 shows a possible approach for DMD gene therapy in humans.
The following were used in the study:

- fibroblasts, which are differentiated cells.
- induced Pluripotent Stem (iPS) cells, which are cells that have been genetically reprogrammed from fibroblast cells.
- an artificial vector that carries the normal *dmd* gene (ART). Using an artificial vector overcomes the size limitations encountered with viral vectors and consequently, allows for the expression of full-length dystrophin protein.

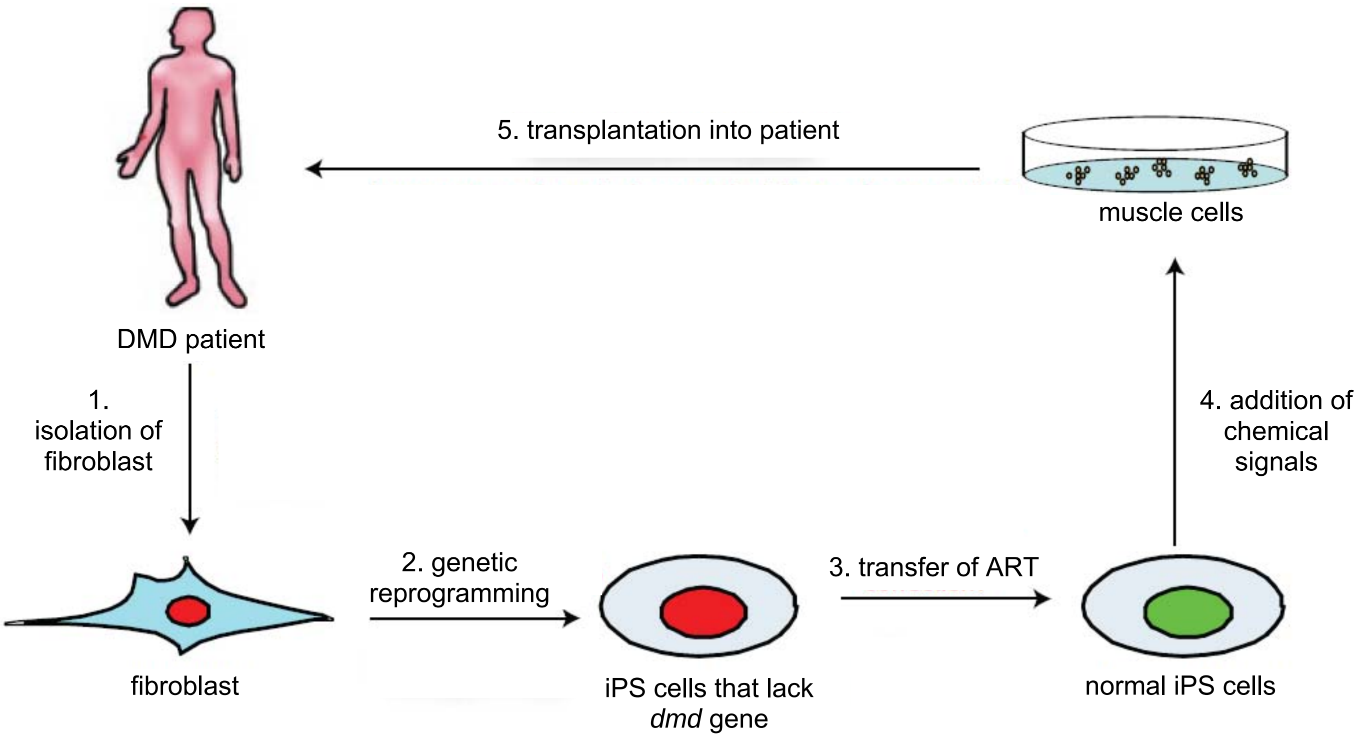

Fig. 2.1

(d) Explain why it is preferable to isolate fibroblast from the the DMD patient.

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(e) Suggest the purpose of adding chemical signals (step 4).

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(f) Upon transplantation into the patient, suggest two disadvantages of this approach that is similar to conventional gene therapy.

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[Total: 12]

QUESTION 3

Many of the original genetically modified crops contained only one modification. An example is Bt Corn A, which produces only one type of Bt protein. To delay Bt resistance in pest populations, such as corn rootworm, modern crops such as Bt Corn B produces two types of Bt proteins against the same pest. Fig. 3.1 shows the survival of corn rootworm larvae on the two types of Bt corn.

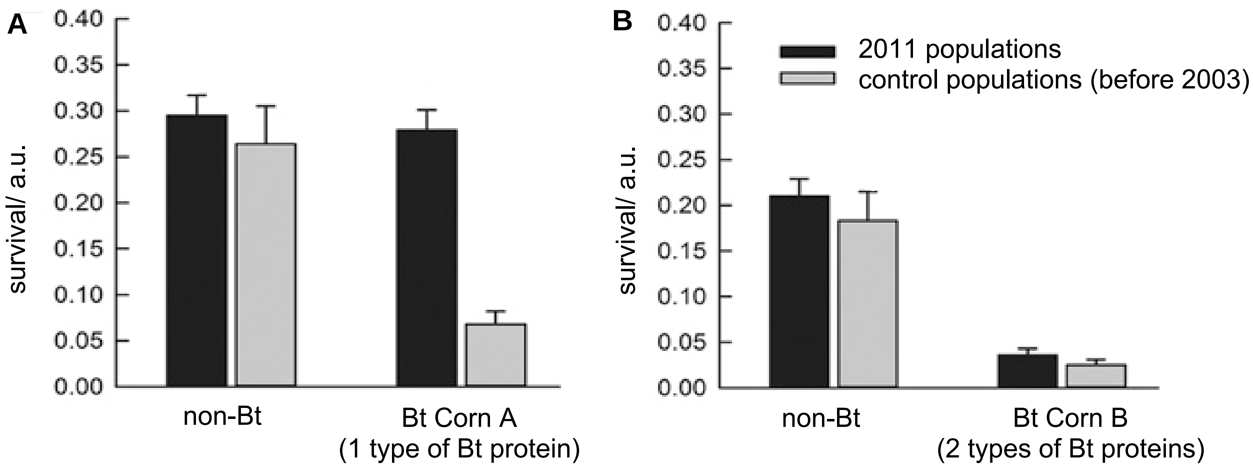

Fig. 3.1

(a) **(i)** With reference to Fig. 3.1A, suggest if there is any significant difference between the 2011 and control populations (before 2003) for non-Bt corn.

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(ii) With reference to Fig. 3.1B, suggest why two types of controls were used.

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(iii) Explain whether expressing one type or two types of Bt proteins is more effective in delaying Bt resistance in corn rootworm.

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Different Bt proteins have varying effects on particular pests. Some Bt proteins are toxic to caterpillars, such as the European corn borer, while other Bt proteins are toxic to rootworms, such as the corn rootworm. Bt genes coding for Bt proteins could be introduced in various combinations, with or without herbicide tolerance genes.

Table 2.1

type of Bt Corn	type of Bt protein	target pest	herbicide tolerance gene
A	Cry1Ab	European corn borer	-
B	Cry3Bb	corn rootworm	-
C	Cry1Ab Cry3Bb	European corn borer corn rootworm	-
D	Cry1Ab Cry3Bb	European corn borer corn rootworm	present

(b) Explain which type of Bt Corn will best increase crop yield.

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.... [2]

Glufosinate ammonium, is a broad-spectrum herbicide, effective in killing a wide range of weed species. Researchers isolated a gene (*bar* gene) derived from soil bacterium *Streptomyces hygroscopicus*, encoding for a herbicide tolerant protein. This gene is introduced to Corn **D** via a plasmid (pCIB3064), resulting in the herbicide tolerant corn.

(c) Outline how herbicide tolerant Corn **D** can be cultured by using calli of Corn **C**.

[5]

(d) Suggest one social issue related to an increase in the use of Corn **D**.

[1]

[Total: 15]

QUESTION 4

Enzymes, such as proteases, are key components found in contact lens cleaning solution. Proteases remove protein deposits from the surfaces of contact lenses.

Two essential contents of such contact lens cleaning solution are:
- pH 7 buffer containing disodium EDTA, a preservative and
- subtilisin A, a protease.

A student researched online and found that there is a range of concentrations of subtilisin A in different commercially available contact lens cleaning solutions. This range of concentration of subtilisin A was between 20 μg cm^{-3} and 100 μg cm^{-3}.

Fig. 4.1 shows how the student simulated a dirty contact lens using protein gelatin and a thin transparent plastic sheet. One side of the plastic sheet was dipped into melted gelatin containing a red dye. The gelatin was then allowed to set and solidify on the plastic sheet, which acts as a support for the gelatin. Subsequently, this simulated dirty contact lens was added to the cleaning solution.

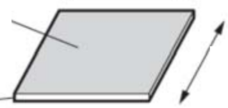

Fig. 4.1

The student tested the activity of subtilisin A by recording the intensity of the red dye in the cleaning solution as shown in Fig. 4.2. This was possible because the cleaning solution was able to remove the red-coloured gelatin from the surface of the transparent plastic sheet.

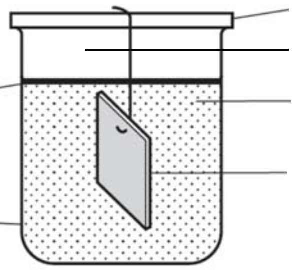

Fig. 4.2

Using this information and your own knowledge, design an experiment to investigate the effect of different concentrations of subtilisin A in commercially available contact lens cleaning solutions on the removal of protein deposits from the simulated dirty contact lenses.

You must use:
- 100 μg cm^{-3} subtilisin A# 

- pH 7 buffer containing disodium EDTA, for dilution purpose
- 12 cm^3 specimen vials, with a wire attached to its lid
- 50 mm x 50 mm thin, transparent plastic sheet coated with red-coloured gelatin
- thermostatically-controlled water bath
- colourimeter and cuvettes

You may select from the following apparatus:

- any normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, graduated pipettes, glass rods etc
- syringes
- white tile
- scalpel
- 15 cm ruler
- timer e.g. stopwatch or stopclock

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- show how you will record your results and the proposed layout of results tables and graphs,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 12]

FREE RESPONSE QUESTION

Your answers must be in continuous prose, where appropriate.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Write your answers in the writing paper provided.
BEGIN EACH PART ON A FRESH SHEET OF WRITING PAPER.
A **NIL RETURN** is required.

QUESTION 5

(a)# Describe the process of polymerase chain reaction. [6]
#

(b)# Explain the advantages and limitations of polymerase chain reaction. [6]
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(c)# Explain the principles of restriction fragment length polymorphism analysis and how it can be used to study genetic variation within a species of organism.# [8]#

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[Total: 20]

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--- END OF PAPER---

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